

The Emerging Science of Interoception: Sensing, Integrating, Interpreting, and Regulating Signals within the Self

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Interoception refers to the representation of the internal states of an organism, and includes the processes by which it senses, interprets, integrates, and regulates signals from within itself. This review presents a unified research framework and attempts to offer definitions for key terms to describe the processes involved in interoception. We elaborate on these definitions through illustrative research findings, and provide brief overviews of central aspects of interoception, including the anatomy and function of neural and non-neural pathways, diseases and disorders, manipulations and interventions, and predictive modeling. We conclude with discussions about major research gaps and challenges.

The Emerging Science of Interoception at the NIH

Neuroscience has progressed tremendously in the past decades in clarifying how we sense and interact with the external world. On the sensory side, this line of research, sometimes referred to as 'exteroception', encompasses (according to most definitions) the primary sensory systems of vision, audition, olfaction, taste, and somatosensation. Less is known about the interoceptive system – the ability of the nervous system to represent our own internal world. On April 16/17 2019, the NIH Blueprint for Neuroscience Research convened a 2 day workshop entitled 'The Science of Interoception and Its Roles in Nervous System Disorders'. At the workshop a distinguished group of investigators highlighted recent findings and discussed a wide range of topics crucial for the future of interoception research.

The workshop identified many critical knowledge gaps in areas related to interoceptive research, including (i) characterization of functional circuits and interaction dynamics between central and peripheral nervous systems in physiological conditions; (ii) delineation of the interaction between interoceptive networks involved in basic physiological processes (e.g., respiration, thirst, feeding, urination, metabolism) and other sensory, motor, reward, emotional, cognitive/memory, and social circuits; (iii) determining the impact of central and peripheral disorders on interoceptive networks, and the effects of modulating interoceptive processes on associated diseases and disorders; and (iv) the need for objective and quantitative assessments of interoception as well as effective technologies and approaches to measure and modulate interoceptive processes.

This article builds on these discussions and efforts to propose a unified framework of interoception science research by defining and describing several key terms. As context for these definitions and for providing some concrete examples of their implications, we also briefly review some key elements of interoceptive processing from the angles of neuroanatomical

Highlights

Interoception refers to the representation of the internal world, and includes the processes by which an organism senses, interprets, integrates, and regulates signals from within itself.

The brain communicates with internal organs via the peripheral nervous system and non-neuronal systems.

Key components of a unified research framework of interoception include interoceptive signals, interoceptors, ascending and descending pathways, central interpreters, central integrators, central regulators, and interoceptive effectors.

In-depth mechanistic studies linking anatomical findings to function are important for defining the roles of the key elements of interoception.

Dysfunction of interoception may be an important component of many neurological, psychiatric, and behavioral disorders.

Better understanding of the neural basis of interoception may provide therapeutic targets for interoceptive dysfunction and related nervous system disorders.

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analysis, function and dynamics, disease implications, potential interventions, computational modeling, and the integration of internal and external representations.

What Is Interoception?

The definition of interoception has evolved over the years. About 150 years ago, the concept was established through the identification of a set of physiological parameters that defined the normal internal state of an organism [1]. In the mid-20th century, the idea evolved to reflect the more dynamic concept of homeostasis [2]. More recently, interoception has been commonly referred to as the process by which the nervous system senses and integrates information about the inner state of the body [3].

Several issues regarding the definition of interoception require careful reevaluation and clarification. First, whereas the terms 'sensing' and 'integrating' seem to imply one-way communication to the brain from other organs, the links between brain and body are often bidirectional and also include communications from the brain to other organs and, in turn, modulation of internal body signals sent back to the brain. Therefore, a more comprehensive definition of interoception should encompass the complex interplay between the brain and other organs that is necessary to monitor and regulate internal states. Second, the anatomical boundary that distinguishes interoceptive and exteroceptive signals requires a nuanced conversation. Conventional wisdom points to the skin as an obvious border, where interoception refers to the processing of signals generated from within the body below the skin. For example, neural activities in subcutaneous tissues, including muscles and connective tissues, that contribute to proprioception, are a form of interoception [4]. Because the anatomical boundaries between interoception and some forms of exteroception become blurred, such as in the case of proprioception and somatosensation, a complementary approach would be to assess whether the signals and the responses of the body represent, rather than originate from, the internal or the external world. Gustation and taste also encounter the dilemma of sensing both the internal and external worlds, but nutrient sensing in the gastrointestinal system is clearly more an indicator of internal status than a representation of the external world. Another example is the vestibular system, that is located in the same sensory organ as the auditory system but typically represents the internal balance of an organism, thus belonging to interoception rather than to exteroception [5].

Given these considerations, we propose a revised description of interoception that may more accurately reflect, we would argue, the bidirectional signal processing between the brain and the internal organs that generates a representation of the internal state of an organism. We recognize that this revised description significantly expands the traditional scope of interoception and may differ from descriptions used in much of the current literature. In this revised description, interoception includes the processes by which an organism senses, interprets, integrates, and regulates signals from within itself. The action of 'sensing' denotes communication from physiological systems outside the CNS to the CNS, through the commonly called ascending pathways, whereas the action of 'regulating' refers to communication from the brain to other physiological systems via descending pathways. The CNS, especially the brain, is primarily responsible for interpreting and integrating these signals into a representation of the internal world. One key difference between this revised definition and some more traditional definitions of interoception is the inclusion of the descending body regulation component. The other key point is that the systems involved in processing signals about the internal environment include not only the peripheral nervous system and the CNS but also components of the vascular, endocrine, and immune systems. In the following sections we describe key concepts related to interoceptive processes and illustrate aspects of the proposed expanded framework, which we hope will foster future avenues of interoception research.

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Interoceptive Signals, Interoceptors, and Sensing Processes

Interoception starts with interoceptive signals originating from within an organism. Interoceptive signals can generally be categorized into three major types. The first includes biochemical signals that range from inorganics, such as acidic ions, to organic molecules and small peptides. The second type includes mechanical forces that alter structures, such as cellular shape, through stretch or tissue extension. The third type includes thermal and electromagnetic signals, which may be delivered in various wave frequencies across the electromagnetic spectrum.

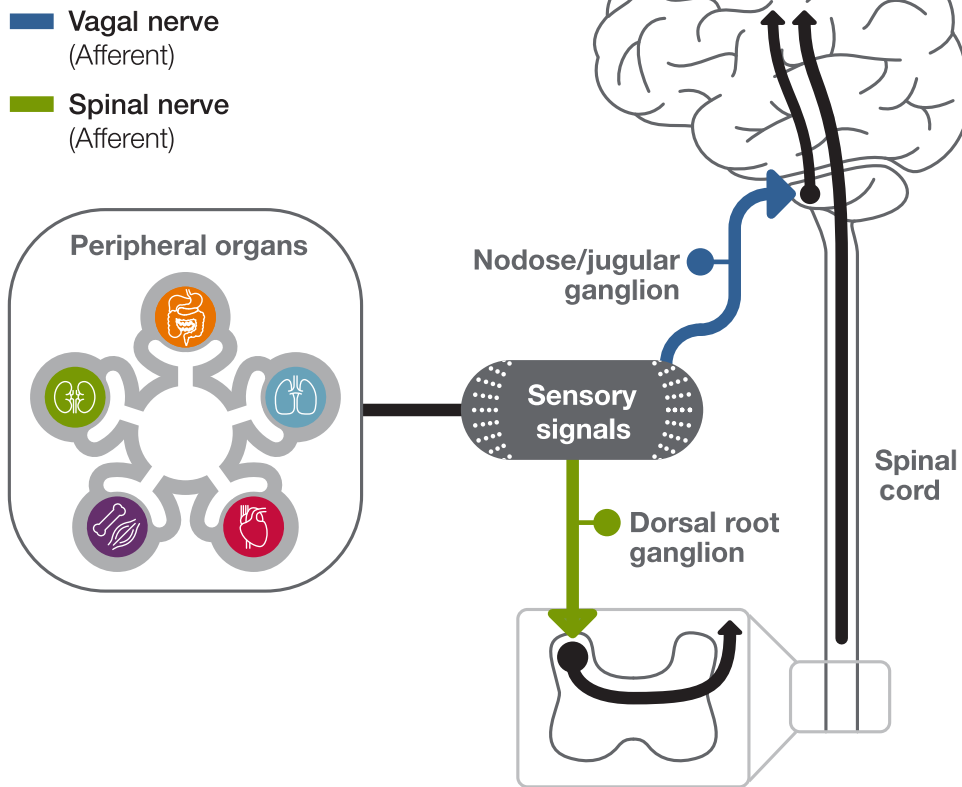
Interoceptors are molecular sensors or receptors in neurons that directly detect these various interoceptive signals and transduce them into electrical, hormonal, or other non-neural signals to be integrated and interpreted by the brain [6]. Interoceptors include chemoreceptors, humoral receptors, specialized mechanoreceptors, and free nerve endings or nociceptors [7]. It is important to note that the biochemical identity of most interoceptors largely remains unknown, and only a few specific examples have been described to date [8,9]. One key challenge for interoception science is the development of systematic approaches to unravel the molecular identities of interoceptors.

Interoceptor location may determine whether the interoceptive signals are transmitted through the peripheral neural system or a non-neural system. For example, chemical interoceptors located on neurons inside the brain most likely receive interoceptive signals through non-neural pathways such as the circulatory or lymphatic systems. Classic neuroendocrine systems such as the hypothalamic–neurohypophyseal system (HNS), the hypothalamic–pituitary–adrenal (HPA) axis, the hypothalamic–pituitary–gonadal axis (HPG), and the hypothalamic–pituitary–thyroid axis (HPT) offer examples of interoception communication via non-neural systems. By contrast, some interoceptors, such as mechanical or thermal interoceptors, are expressed in peripheral nerve termini and directly detect signals in local organs, induce the peripheral sensory ganglia to generate electrical signals, and thereby transmit the interoceptive information through the peripheral neural pathways to the brain (Figure 1).

There are generally two major ascending peripheral neural or afferent pathways that transmit interoceptive signals to the CNS [10–13]. Signals in these two pathways commonly relay through two distinct types of peripheral sensory ganglia. Ganglia residing in the cranial/vagal pathways, such as nodose or jugular ganglia, often project to the nucleus tractus solitarius (NTS) of the brainstem, whereas dorsal root ganglia, located along the spinal nerve pathway, project information to the brain through the spinal cord [13]. Visceral afferents that travel along cranial nerves, including the vagus, can also be referred to as 'parasympathetic afferents', whereas those that travel through the dorsal column of the spinal cord are often called 'sympathetic afferents' [10]. It is hypothesized that vagal afferents primarily carry mechanoreceptor and chemosensory signals; spinal afferents carry signals related to temperature, pain, and tissue injury [12–15]. Some evidence has suggested that vagal and spinal afferents may represent opposing parasympathetic and sympathetic signals, and may thus interact within the interoceptive regions of the brain to inhibit each other [13]. However, much remains to be studied to assess the differences between these two types of ascending neural pathways and their impact.

Central interpreters and integrators of interoception include neurons in the CNS involved in processing, interpreting, and/or integrating interoceptive signals. Whether delivered through the humoral, lymphatic, or peripheral nervous systems, interoceptive information is often first processed in subcortical structures of the brain such as the medial NTS, the parabrachial nucleus (PB), and the ventromedial nucleus of the thalamus [4,11,12,16,17] (Figure 2). In turn, the neurons in these structures may project to higher brain regions including the hypothalamus, insula,

Ascending neural pathways

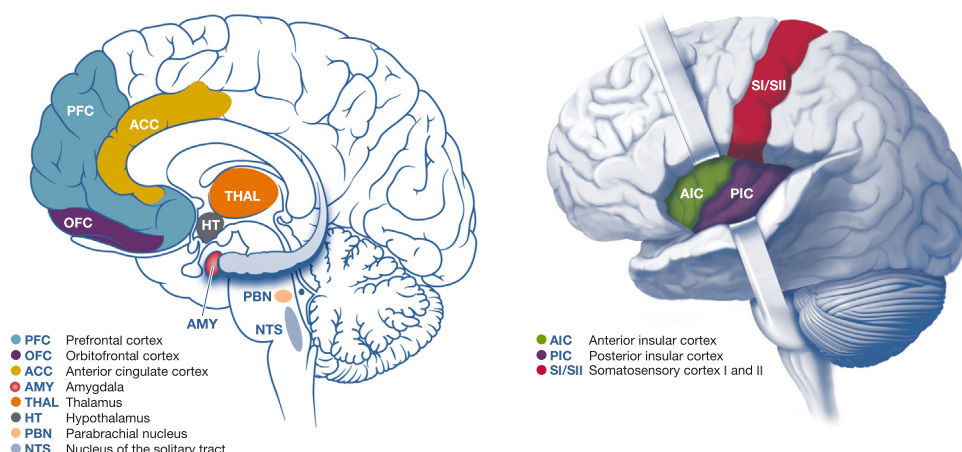


Trends in Neurosciences

Figure 1. Illustrative Diagram of Sample Ascending Neural Pathways of Interoception. This diagram only describes the ascending neural pathways connecting the peripheral internal organs to the brain. The non-neural ascending pathways are not included in this figure. Interoceptive sensory signals generated by the peripheral organs may be detected by molecular sensors or receptors, termed interoceptors, that are present at the termini of the peripheral sensory ganglia. The sensory ganglia residing in the cranial/vagal pathways, such as nodose or jugular ganglia, often send projections to the nucleus tractus solitarius (NTS) of the brainstem through vagal afferents, sometimes referred to as 'parasympathetic afferents'. The sensory ganglia located along the spinal nerve pathway, often called dorsal root ganglia, send projections through the dorsal column of the spinal cord into the brain via spinal afferents, sometimes referred to as 'sympathetic afferents'.

anterior cingulate cortex, and somatosensory cortex for further integration and interpretation [5,18–24].

Early in the history of the field, the insula emerged as a crucial cortical node in the interoceptive system. Penfield's neurosurgical stimulation experiments first connected insular cortex to visceral sensation, and neuroanatomical analyses confirmed a viscerotopic map in the insula [12,13,16,25–27]. Although our understanding of the anatomical and functional parcellation of the insula remains incomplete, studies in mammals ranging from rodents to humans consistently reveal a posterior-to-anterior topography in the insular interoceptive map [18,27–29]. Primary interoceptive information is relayed from the ventromedial nucleus of the thalamus to the posterior insula, and integration with exteroceptive sensorimotor and proprioceptive information most likely takes place within the posterior and central regions [22,30]. The anterior insular cortex (AIC) is



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Figure 2. Brain Regions Involved in Central Processing of Interoceptive Signals. The diagrams in this figure depict some of the various brain regions involved in central interpretation, integration, and regulation of interoceptive information. The figure on the left is a mid-sagittal view of the human brain, highlighting brain regions implicated in central processing of interoception. The figure on the right is a lateral/sagittal view of the brain highlighting the insula cortex, including anterior insular cortex (AIC) and posterior insular cortex (PIC), as well as somatosensory cortex (SI/SII). Within the brain, interoceptive information is often first processed in subcortical structures such as the medial nucleus of the solitary tract (NTS), the parabrachial nucleus (PB), and then the ventromedial nucleus of the thalamus. Neurons in these structures project to other brain regions such as the hypothalamus and eventually to the insular cortex, a crucial cortical node in the interoceptive system. Primary interoceptive information relayed from the ventromedial nucleus of the thalamus projects to PIC, where integration with exteroceptive sensorimotor and proprioceptive information from SI/II most likely occurs. AIC is most strongly connected to paralimbic cortical regions such as the orbitofrontal cortex (OFC), anterior cingulate cortex (ACC), and prefrontal cortex (PFC), which may be involved in linking interoceptive with emotional or cognitive states for further integration and interpretation, and possibly in generating regulatory signals that are sent back to lower brain regions involved in interoceptive descending efferent systems.

most strongly connected to paralimbic cortical regions such as the orbitofrontal cortex (OFC) and the anterior cingulate cortex (ACC), and may be involved in connections between interoceptive and emotional or cognitive states [4,16,24,26,29,31–33]. It is important to note some key anatomical differences in the insula across species [34]. The uniquely shaped and functionally mysterious layer 5 von Economo and fork neurons are found primarily in the AIC and ACC of macaques, great apes, and humans, as well as in several other large and highly social mammalian species such as elephants and whales [26,29,35]. The insula is considered to be part of the ventral salience network for arousal-based affective experience. The ventral salience network is largely homologous between monkeys and humans, but the dorsal network for attentional control is much more developed in humans [27,30,33,36–38]. The insula also receives major inputs from the amygdala, but rodents and monkeys have markedly different organizations of amygdalar nuclei [39,40].

Interoceptive information may not require higher levels of cortical processing. In other cases, interoceptive signals may engage higher-order processing at perceptual, cognitive, and/or affective levels, thus rising to the level of conscious awareness [6]. In humans, the insula is activated when individuals consciously attend to their own interoceptive states, suggesting that it may serve as a key interoceptive hub for integrating and regulating signals from the internal and external environments [26]. Understanding how and when autonomic versus conscious processing of interoceptive information occurs remains an intriguing area for further in-depth investigations [7].

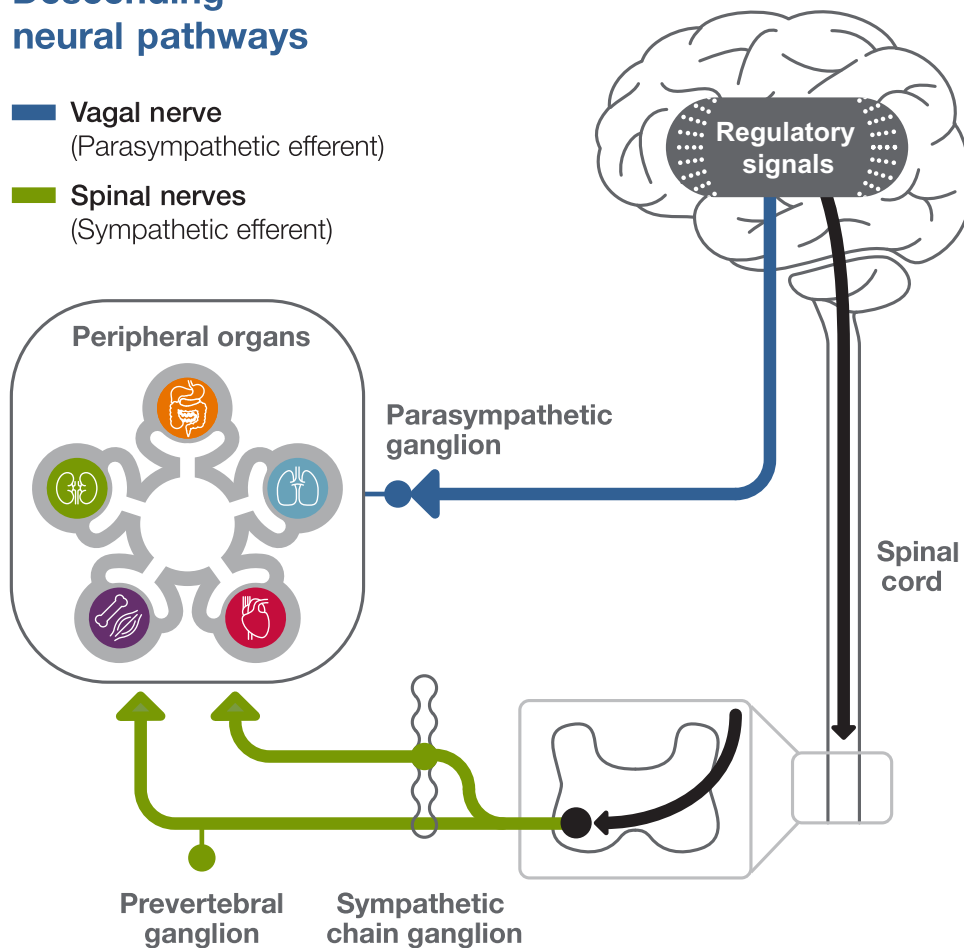
Regulation of Interoception

The concept of regulating interoception via descending pathways has not been incorporated into most definitions of interoception, despite the well-established ability of the CNS to generate

signals that regulate the internal state [41,42]. We will refer to these signals generated by the CNS to regulate the interoceptive processes, often in response to interoceptive input or cognitive and exteroceptive factors, as 'regulatory signals of interoception'. Neurons in the CNS involved in generating these regulatory signals can be called central regulators of interoception (Figure 3).

Descending neural pathways

- **Vagal nerve**
(Parasympathetic efferent)
- **Spinal nerves**
(Sympathetic efferent)



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Figure 3. Illustrative Diagram of Sample Descending Neural Pathways of Interoception. This diagram only describes the descending neural pathways connecting the brain to the peripheral internal organs. The non-neural descending pathways are not included in this figure. We refer to the signals generated by the central nervous system (CNS), often in response to interoceptive input, to regulate the interoceptive processes as 'regulatory signals of interoception'. Neurons in the CNS involved in generating these regulatory signals can be called central regulators of interoception. The regulatory signals may be communicated to the peripheral internal organs via descending spinal or vagal/cranial efferents. The spinal efferents are also referred to as the sympathetic efferents of the autonomic nervous system because they go through the spinal cord pathway and regulate sympathetic activities. After exiting the spinal cord, some of the spinal efferents synapse onto the sympathetic chain ganglia, which then directly project onto peripheral organ cells to regulate interoceptive signals or organ function. These sympathetic chain ganglia may thus be considered to be the effector neurons in the interoceptive descending neural pathway. Other spinal efferents synapse instead onto prevertebral ganglia, and these which may act as effector neurons and project to the peripheral organ cells. The vagal/cranial efferents, also called the parasympathetic efferents of the autonomic nervous system, typically go through the nodose or jugular ganglia bundles and make synaptic connections onto parasympathetic ganglia, which are often located near the peripheral organs and act as effector neurons to regulate peripheral organ interoceptive signals or organ functions.

Regulatory signals of interoception, similar to interoceptive signals, can be transmitted via both non-neural (e.g., humoral) and neural pathways (e.g., cranial/vagal or spinal efferents) that target the peripheral organs [23,43]. In non-neural pathways, regulatory signals are delivered to the peripheral organ via the vascular or lymphatic systems and interact directly with the responding non-neural cells. In neural pathways, the final effectors, commonly called sympathetic or parasympathetic ganglion neurons, receive input from the brain through the spinal or vagal/cranial efferent nerves and directly synapse with internal organ non-neural cells [43] (Figure 3). It is worth noting that the non-neural and neural pathways may interact, such that a regulatory signal may be initially delivered through one pathway (e.g., a non-neural pathway) to impact on the other pathway (e.g., neural pathway) before reaching its final effector target.

The primary function of the regulatory signals of interoception is to regulate the generation and transmission of interoceptive signals of the targeted internal organs, which can be called effectors, and thus complete the circle of interoceptive processes. However, often the impact of the regulatory signals may be best measured by the responses or changes of function in the target organs. It is therefore sometimes impossible to distinguish regulatory interoceptive signals from body regulation. In our view, the inclusion of body regulation as a component of interoception science not only is necessary from a neuroanatomical perspective but also enables the development of more innovative methods to probe the functional impact of interoceptive processes, although we recognize that, in the current literature, many have described body regulation and interoception as separate terms and concepts.

Gaps and Challenges of Research in Interoceptive Neural Circuits

The picture of the interoceptive nervous system sketched in the previous text (Figures 1–3) is certainly incomplete. There are many gaps and methodological challenges in studying neural circuits of interoception at the neuroanatomical and functional levels.

The neuroanatomical techniques used to establish the knowledge base of interoception have primarily included histochemical and autoradiographic tract tracing, cytoarchitectonic analyses, and magnetic resonance imaging (MRI)-based diffusion tractography (DTI) [13,31,44,45]. Notably, tract-tracing studies show highly collateralized and interconnected networks across all nodes from the periphery to the cortex [22,46]. Additional experiments using tools such as transneuronal tracers and virus-based techniques will be necessary to establish a more detailed picture [17,22]. There are also multiple gaps between research focused on inputs from the periphery to the brainstem and those focused on the insular cortex [47–49]. For instance, more attention needs to be paid to examining the thalamic relay(s) [45]. In addition, much remains to be understood regarding the connections between the ascending and descending pathways, such as the hypothalamic endocrine connections and the connections from insula to other brain regions including somatosensory cortex and beyond. How do the neuronal and non-neural pathways connecting brain and periphery interact and influence each other [16]?

Converging and diverging projections are known to exist at all levels of interoception, but crucial cellular and synaptic information is missing. Experiments focused on the molecular-level specificity and diversity of neuronal types within the interoceptive system have only begun to emerge [17,50–52]. For instance, although attention has been paid to the different cytoarchitectonic regions of the insular cortex, is there a single representation of the body in the insula, or might there be multiple, overlapping maps [26,52]? Furthermore, it is important to note that the properties and cell types of the insular cortex appear to differ substantially across species, limiting the applicability of some findings from rodents to human interoceptive health [34]. At the periphery, transneuronal tracers have had limited application to date, consisting

primarily of anterograde studies from scattered visceral organ systems [22,46,53–55]. Visceral afferents are low-density, thin, unmyelinated fibers that can be difficult to visualize [56]. Focusing on cell-type analysis and imaging of regions or nodes with a high density of neurons, such as various types of peripheral sensory ganglia, might offer an excellent opportunity to gain more comprehensive understanding [56–62]. Moreover, little is known about the diverse cell types involved in many of the key information-processing nodes for interoception from peripheral ganglia to cortex, whereas the complexity of circuits within these nodes, such as the brainstem and hypothalamus, remains largely unmapped.

Functional techniques have been mostly limited to early studies of evoked potentials in vagotomized animals and very few human lesion case reports. A more recent collection of resting state and task-based functional MRI (fMRI) studies in humans have revealed widespread viscera–brain coupling and placed insula function within the larger 'salience network' [27,30,33,36–38,63]. The insula and associated cortical and subcortical networks have been correlated with a wide range of possible functions related to interoception and allostasis, including visual perception, mental time-keeping, emotion, empathy, language and music perception, and self-awareness [23,26,27,34,38]. However, much remains to be understood about the many brain regions involved in interoception, especially whether specific neuronal populations in these regions function as interpreters, integrators, or regulators of interoceptive information.

Although interoception studies in humans currently rely primarily on fMRI blood oxygen level-dependent (BOLD) and DTI structural imaging [16] to provide crucial information about large-scale function, these methodologies are fundamentally correlative. In-depth, mechanistic studies that link anatomical findings to function will be important to substantiate the roles of brain regions and domains in interoception. In human subject research, non-invasive methodologies such as transcranial magnetic stimulation (TMS) and transcranial direct current stimulation (tDCS) may offer opportunities to assess the causal effects of brain regions in interoception [64]. However, the deep location of some of the brain regions involved in interoception, including the insular cortex, makes them difficult to image and stimulate [33]. Imaging studies in lesion patients, for instance AIC lesions, may also offer some causal evidence although they require caution in interpretation [24].

Functional Impact and Dynamics

Optimal sensing, interpretation, integration, and regulation of internal body signals, whether rising to the level of conscious awareness or not, are crucial for many essential physiological functions, such as breathing, eating, drinking, micturition, and maintaining body temperature, as well as for psychological experiences ranging from a variety of feelings and emotions to motivations and adaptive behaviors [3]. In addition, interoceptive and exteroceptive processes may interact to orchestrate complex physiological and behavioral functions, such as in the stress response [65]. The functions of interoception may range from essential bodily functions to high-level cognitive and emotional behaviors [66], and can be roughly grouped into two categories based on the direction of information flow.

The first category of interoceptive functions includes information flow from the body to the brain. For example, it is mostly unknown how affect – a brain state that can be described as a low-dimensional representation of high-dimensional behavioral state in the body [67] – is influenced by interoception. However, recent work on gut and hippocampal function [17], which tested the hypothesis that specific interoceptive signals are transported via the vagus nerve to the hippocampus to influence memory function, provides an elegant example on how one could begin to address this challenge.

The second category of interoceptive functions includes information flow from the brain to the body to exert bodily effects. The voluntary urination model is a good example. The target output of voluntary urination is the urethral sphincter, which is somatically controlled by the pontine micturition center (PMC, or Barrington's nucleus) of the brain [68], particularly corticotropin-releasing factor (CRF) labeled subpopulations of neurons in the PMC [69,70]. Stimulation of CRF-labeled neurons in the PMC caused immediate urination in mice [69], suggesting that the PMC neurons are sufficient to drive behavior. When neural activity was blocked in CRF-labeled neurons in the PMC, voluntary urination was inhibited. Often, this category of interoceptive function utilizes autonomic nervous system pathways, either sympathetic or parasympathetic.

In animal models, it is often possible to couple (i) neural activity manipulations, even at the single neuron level; (ii) neural activity measurements, such as electrophysiological or physiological assays, and/or *in vivo* neural imaging; with (iii) functional assessments to measure organ physiology and/or behavioral assays. Such experimental combinations have the potential to best support rigorous gain- and loss-of-function analyses of interoceptive neural circuits. By contrast, assessments of interoception in humans have been largely limited to a handful of approaches and remain mostly correlational. These approaches include heart-beat measures [71–73], skin conductance responses [74,75], and subjective self-report measures such as interoceptive accuracy or awareness [76]. More recently, microneurography has emerged as an objective tool to probe interoceptive responses [77,78]. If one includes body regulation as part of interoception research, additional physiological measures, such as blood pressure and baroreflex [79], may be explored to assess the functional impact of interoception. A key caveat in this context is that many organ functions are vital, making causal analyses challenging in human subject research of interoception. Furthermore, developing consistent experimental measures and metrics for comparable studies in humans and animal models will be crucial for translating insights across animal model and human studies.

Dynamic changes in interoceptive pathways over the lifespan remain largely unexplored. Several research groups have begun to study interoception in early life [80] and to examine how interoception changes with age [81–83], but relatively little is known about lifelong changes in interoception. One key challenge is the lack of a measure of interoceptive sensitivity that is suitable for non-verbal populations such as infants [80] or older adults with dementia [84]. In addition, conducting longitudinal studies and identifying milestones of interoception development will be necessary to uncover the changes in interoception across the lifespan.

Diseases and Disorders

Dysfunction of interoception has been increasingly recognized as an important component of many neurological, psychiatric, and behavioral disorders [3]. Altered structure, functional activity, or connectivity within the interoceptive neural network has been observed for instance in individuals with migraine and other types of chronic pain [85,86], alcohol and substance use disorders [87], anxiety, depression, and affective disorders [88,89], post-traumatic stress disorder [90], obsessive compulsive disorder [91], autism spectrum disorder (ASD) [92], eating disorders [93], somatic symptom disorders [94], stroke and neurodegenerative diseases [95,96], and may be a basis for comorbidity of neuropsychiatric and mental disorders [3,97].

For example, irritable bowel syndrome, characterized by visceral hypersensitivity, is associated with disruptions in the endocrine and immune systems in the gut as well as changes in the cortical neuronal network of sensorimotor, salience, emotion, and arousal [54] and disrupted modulation of insular cortex [98] in the human brain. Similarly, obesity is associated with disrupted interoceptive states from the gut to the brain such as altered functional connectivity of the insula [99] and

distributed brain regions between the dorsal mid-insula, medial OFC, dorsal striatum, and ventral striatum [100].

Many psychiatric and neurodevelopmental disorders, such as schizophrenia [101], attention-deficit hyperactivity disorder [102], ASD, depression, and anxiety disorders, are not only linked to altered brain networks that are crucial for the integration of interoceptive signals for emotion processing and cognition but also exhibit a variety of physical symptoms [3]. For example, ASD is associated with dysregulated anterior insula connectivity and deficient emotional processing that may be due to aberrant prediction errors in interoceptive processing [72,103]. At the same time, individuals with ASD often show interoceptive changes such as increased sensitivity to pain and gastrointestinal symptoms [104].

Similarly, addiction to alcohol or other substances of abuse may cause adaptation in the interoceptive network which not only exacerbates stress and contributes to the alteration of emotion and reward processing but also gives rise to comorbid conditions through central and peripheral interactions [105,106]. A recent animal model study identified a genetic factor that links the neurocircuit adaptation of tobacco smoking to type 2 diabetes in humans [107], and provided evidence to show that the brain can control peripheral organ function while peripheral organs can regulate the addictive properties of nicotine. The gut–brain connection has also been shown to play a role in addictive behaviors and alcohol use disorders. For instance, ghrelin is a hunger hormone secreted in the stomach, and its concentration in individuals with alcohol use disorder correlates with alcohol drinking and predicts alcohol relapse [108,109]. In rodents, administration of ghrelin leads to an increase in alcohol reward and in alcohol intake, administration, and preference, whereas blockade of the ghrelin receptor resulted in reduction of these behaviors [108,109]. Moreover, evidence suggests that ghrelin, delivered through the vascular system, may mediate signaling from gut to brain via the vagus nerve [17,110].

A wide range of diseases and disorders are related to interoception [3,97], and pathophysiological and causal evidence has been generated for some of these conditions. However, it is important to note that, in most of them, the links to interoceptive dysregulation are largely symptomatic and descriptive. Future pathophysiological studies in clinical populations as well as in appropriate animal models are necessary for deepening the understanding of the possible outcomes of dysregulated interoceptive processing.

Manipulations and Interventions

The complex interplay between the ascending and descending pathways of the interoceptive system provides many potential routes and methods for targeted interventions in interoceptive dysfunction and related disorders. A multitude of approaches, ranging from non-invasive behavioral manipulations to specific pharmacological and neural stimulation interventions, have been considered for regulatory interoceptive processes [111].

In general, these approaches can be categorized into three groups – behavioral, neural stimulation, and pharmacological – each with advantages and limitations. Behavioral interventions, such as meditation and cognitive behavioral therapy (CBT), are the least invasive and generally the safest of the various interventions [112]. These approaches utilize exteroceptive routes (e.g., sound, vision, somatosensation, and cognitive influence) to trigger activity in the brain, and likely exert their effects on interoceptive body signals and function [113] through descending pathways. The limitations of these approaches include a lack of established therapeutic targets, relatively low potency, and delayed improvements [114]. Current neural stimulation approaches include TMS, tDCS, deep brain stimulation (DBS), vagal nerve stimulation (VNS), and transcutaneous electrical

nerve stimulation (TENS) targeting peripheral nerves [115]. If the therapeutic targets are clear and specific, nerve stimulation can be a potent therapeutic with few side effects. However, therapeutic targets for most of these approaches have yet to be identified and validated. In addition, some approaches such as DBS and VNS may require invasive neurosurgery. Pharmacological interventions, such as blockade of the ghrelin receptor and/or of the peptide itself, represent promising approaches, although additional translational work is needed in that regard [108,109]. The pharmacological approach could be particularly effective for targeting interoceptors either within the brain, at the periphery, or along associated molecular signaling pathways. As in most pharmacological approaches, off-target effects and related side effects pose major challenges.

Regardless of the type of interventional approaches, identifying, assessing, and validating the therapeutic targets through rigorous mechanistic studies in both humans and appropriate animal models will be crucial for safety and efficacy in treating interoception-related diseases and disorders. In addition, group and individual variables including sex, gender, age, and social factors are important elements to consider for interventional studies.

Predictive Modeling

Computational modeling is an essential component of interoception research because it helps to formalize the empirical findings within a mathematical framework, as well as providing predictions for future experiments [116–118]. Several mathematical models are currently used to describe both the perceptual and body regulation aspects of interoception [117,119], although the sensory or perceptual aspects of interoception and the body regulation aspect are often modeled as separate processes. It may be desirable to develop computational models that could capture and describe the proposed framework of interoception in this article. In addition, modeling how interoceptive processes may be integrated with exteroceptive processes can provide a more comprehensive picture of how our nervous system interacts with the rest of the body to maintain function and support survival [119]. Modeling key parameters and outcomes that are accessible in experimental studies, such as 'interoceptive accuracy' and 'interoceptive awareness' [72,76], can greatly facilitate the development and validation of computational models and should be strongly encouraged.

Concluding Remarks

Although the foundations of the science of interoception were laid over 100 years ago, interoception research mostly regained momentum in recent years, partly because of the availability of high-resolution, multimodal tools for interrogating interoceptive processes. In this article we have outlined a proposed comprehensive framework of interoception science which may help to accelerate progress towards an integrative understanding of how we sense and regulate our internal states. Both conceptual and technical/methodological challenges remain (see [Outstanding Questions](#)), and our hope is that this review, together with other articles in this Special Issue, will offer stimulating ideas to enrich the emerging science of interoception.

Disclaimer Statement

The content is solely the responsibility of the authors and does not represent the official views of the NIH or federal government.

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Outstanding Questions

Interoceptors are molecular sensors or receptors in neurons that directly detect various interoceptive signals and transduce them into electrical, hormonal, or other non-neural signals to be integrated and interpreted by the brain. What are the molecular entities of these interoceptors? How can one systematically identify them?

What are the functional circuits of interoception? How do the central and peripheral nervous systems dynamically interact to support interoceptive processing in both physiological and pathological conditions?

What is the impact of central or peripheral disorders on interoceptive networks?

What are the effects of modulating interoceptive processes on diseases and disorders of the nervous system?

How can one develop more objective and quantitative assessments of interoception?

What are the most effective strategies for developing technologies and approaches to modulate interoceptive processes?

How do the interoceptive networks interact with other sensory, motor, reward, emotional, and cognitive/memory circuits to regulate nervous system diseases and disorders?

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References

- Cameron, O.G. (2001) Interoception: the inside story – a model for psychosomatic processes. *Psychosom. Med.* 63, 697–710
- Wiener, N. (1953) The concept of homeostasis in medicine. *Trans. Stud. Coll. Physicians Phila* 20, 87–93
- Khalsa, S.S. *et al.* (2018) Interoception and mental health: a roadmap. *Biol. Psychiatry Cogn. Neurosci. Neuroimaging* 3, 501–513
- Tuthill, J.C. and Azim, E. (2018) Proprioception. *Curr. Biol.* 28, R194–R203
- Shinder, M.E. and Newlands, S.D. (2014) Sensory convergence in the parieto-insular vestibular cortex. *J. Neurophysiol.* 111, 2445–2464
- Denton, D. *et al.* (1999) Neuroimaging of genesis and satiation of thirst and an interoceptor-driven theory of origins of primary consciousness. *Proc. Natl. Acad. Sci. U. S. A.* 96, 5304–5309
- Berntson, G.G. and Khalsa, S.S. (2021) Neural circuits of interoception. *Trends Neurosci.* 44, 17–28
- Ranade, S.S. *et al.* (2014) Piezo2 is the major transducer of mechanical forces for touch sensation in mice. *Nature* 516, 121–125
- Chesler, A.T. *et al.* (2016) The role of PIEZO2 in human mechanosensation. *N. Engl. J. Med.* 375, 1355–1364
- Mei, N. (1983) Recent studies on intestinal vagal afferent innervation. Functional implications. *J. Auton. Nerv. Syst.* 9, 199–206
- Janig, W. (1996) Neurobiology of visceral afferent neurons: neuroanatomy, functions, organ regulations and sensations. *Biol. Psychol.* 42, 29–51
- Saper, C.B. (2002) The central autonomic nervous system: conscious visceral perception and autonomic pattern generation. *Annu. Rev. Neurosci.* 25, 433–469
- Craig, A.D. (2002) How do you feel? Interoception: the sense of the physiological condition of the body. *Nat. Rev. Neurosci.* 3, 655–666
- Barone, F.C. and Zarco, I. (1995) de Coronado, and M.J. Wayner, Gastric distension modulates hypothalamic neurons via a sympathetic afferent path through the mesencephalic periaqueductal gray. *Brain Res. Bull.* 38, 239–251
- Taher, J. *et al.* (2017) Central nervous system regulation of hepatic lipid and lipoprotein metabolism. *Curr. Opin. Lipidol.* 28, 32–38
- Critchley, H.D. and Harrison, N.A. (2013) Visceral influences on brain and behavior. *Neuron* 77, 624–638
- Suarez, A.N. *et al.* (2018) Gut vagal sensory signaling regulates hippocampus function through multi-order pathways. *Nat. Commun.* 9, 2181
- Cechetto, D.F. and Saper, C.B. (1987) Evidence for a viscerotopic sensory representation in the cortex and thalamus in the rat. *J. Comp. Neurol.* 262, 27–45
- Cobos, I. and Seeley, W.W. (2015) Human von Economo neurons express transcription factors associated with layer V subcortical projection neurons. *Cereb. Cortex* 25, 213–220
- Champagne, D. *et al.* (1998) CRFergic innervation of the paraventricular nucleus of the rat hypothalamus: a tract-tracing study. *J. Neuroendocrinol.* 10, 119–131
- Iwai, H. *et al.* (2015) Ascending parabrachio-thalamo-striatal pathways: potential circuits for integration of gustatory and oral motor functions. *Neuroscience* 294, 1–13
- Dum, R.P. *et al.* (2009) The spinothalamic system targets motor and sensory areas in the cerebral cortex of monkeys. *J. Neurosci.* 29, 14223–14235
- Gu, X. *et al.* (2013) Anterior insular cortex and emotional awareness. *J. Comp. Neurol.* 521, 3371–3388
- Wang, X. *et al.* (2019) Anterior insular cortex plays a critical role in interoceptive attention. *Elife* 8, e42265
- Penfield, W. and Faulk Jr., M.E. (1955) The insula; further observations on its function. *Brain* 78, 445–470
- Craig, A.D. (2009) How do you feel – now? The anterior insula and human awareness. *Nat. Rev. Neurosci.* 10, 59–70
- Azzalini, D. *et al.* (2019) Visceral signals shape brain dynamics and cognition. *Trends Cogn. Sci.* 23, 488–509
- Kelly, C. *et al.* (2012) A convergent functional architecture of the insula emerges across imaging modalities. *Neuroimage* 61, 1129–1142
- Gogolla, N. (2017) The insular cortex. *Curr. Biol.* 27, R580–R586
- King, A.B. *et al.* (1999) Human forebrain activation by visceral stimuli. *J. Comp. Neurol.* 413, 572–582
- Mufson, E.J. and Mesulam, M.M. (1982) Insula of the old world monkey. II. Afferent cortical input and comments on the claustrum. *J. Comp. Neurol.* 212, 23–37
- Khalsa, S.S. *et al.* (2009) The pathways of interoceptive awareness. *Nat. Neurosci.* 12, 1494–1496
- Uddin, L.Q. *et al.* (2017) Structure and function of the human insula. *J. Clin. Neurophysiol.* 34, 300–306
- Butti, C. and Hof, P.R. (2010) The insular cortex: a comparative perspective. *Brain Struct. Funct.* 214, 477–493
- Evrard, H.C. *et al.* (2012) Von Economo neurons in the anterior insula of the macaque monkey. *Neuron* 74, 482–489
- Seeley, W.W. *et al.* (2007) Dissociable intrinsic connectivity networks for salience processing and executive control. *J. Neurosci.* 27, 2349–2356
- Kurth, F. *et al.* (2010) A link between the systems: functional differentiation and integration within the human insula revealed by meta-analysis. *Brain Struct. Funct.* 214, 519–534
- Kleckner, I.R. *et al.* (2017) Evidence for a large-scale brain system supporting allostasis and interoception in humans. *Nat. Hum. Behav.* 1, 0069
- Mesulam, M.M. and Mufson, E.J. (1982) Insula of the old world monkey. III. Efferent cortical output and comments on function. *J. Comp. Neurol.* 212, 38–52
- Swanson, L.W. and Petrovich, G.D. (1998) What is the amygdala? *Trends Neurosci.* 21, 323–331
- Li, M.M. *et al.* (2019) The paraventricular hypothalamus regulates satiety and prevents obesity via two genetically distinct circuits. *Neuron* 102, 653–667
- Taggart, P. *et al.* (2016) Significance of neuro-cardiac control mechanisms governed by higher regions of the brain. *Auton. Neurosci.* 199, 54–65
- Janig, W. and Habler, H.J. (2000) Specificity in the organization of the autonomic nervous system: a basis for precise neural regulation of homeostatic and protective body functions. *Prog. Brain Res.* 122, 351–367
- Mesulam, M.M. and Mufson, E.J. (1982) Insula of the old world monkey. I. Architectonics in the insulo-orbito-temporal component of the paralimbic brain. *J. Comp. Neurol.* 212, 1–22
- Mufson, E.J. and Mesulam, M.M. (1984) Thalamic connections of the insula in the rhesus monkey and comments on the paralimbic connectivity of the medial pulvinar nucleus. *J. Comp. Neurol.* 227, 109–120
- Dum, R.P. *et al.* (2019) The mind–body problem: circuits that link the cerebral cortex to the adrenal medulla. *Proc. Natl. Acad. Sci. U. S. A.* 116, 26321–26328
- Ruggiero, D.A. *et al.* (1987) A role of insular cortex in cardiovascular function. *J. Comp. Neurol.* 257, 189–207
- Gehrlach, D.A. *et al.* (2019) Aversive state processing in the posterior insular cortex. *Nat. Neurosci.* 22, 1424–1437
- Livneh, Y. *et al.* (2020) Estimation of current and future physiological states in insular cortex. *Neuron* 105, 1094–1111

50. Kaelin-Langer, M.M. *et al.* (2018) A gut-brain neural circuit for nutrient sensory transduction. *Science* 361, 26321–26328
51. Yang, L. *et al.* (2019) Transcriptomic landscape of von Economo neurons in human anterior cingulate cortex revealed by microdissected-cell RNA sequencing. *Cereb. Cortex* 29, 838–851
52. Evrard, H.C. (2019) The organization of the primate insular cortex. *Front. Neuroanat.* 13, 43
53. Rinaman, L. and Schwartz, G. (2004) Anterograde transneuronal viral tracing of central viscerosensory pathways in rats. *J. Neurosci.* 24, 2782–2786
54. Rinaman, L. (2010) Ascending projections from the caudal visceral nucleus of the solitary tract to brain regions involved in food intake and energy expenditure. *Brain Res.* 1350, 18–34
55. McGovern, A.E. *et al.* (2012) Transneuronal tracing of airways-related sensory circuitry using herpes simplex virus 1, strain H129. *Neuroscience* 207, 148–166
56. Wang, W. (2018) Optogenetic manipulation of ENS – the brain in the gut. *Life Sci.* 192, 18–25
57. Udit, S. and Gautron, L. (2013) Molecular anatomy of the gut-brain axis revealed with transgenic technologies: implications in metabolic research. *Front. Neurosci.* 7, 134
58. Kupari, J. *et al.* (2019) An atlas of vagal sensory neurons and their molecular specialization. *Cell Rep.* 27, 2508–2523
59. Chang, R.B. *et al.* (2015) Vagal sensory neuron subtypes that differentially control breathing. *Cell* 161, 622–633
60. Williams, E.K. *et al.* (2016) Sensory neurons that detect stretch and nutrients in the digestive system. *Cell* 166, 209–221
61. Han, W. *et al.* (2018) A neural circuit for gut-induced reward. *Cell* 175, 665–678 e23
62. Rajendran, P.S. *et al.* (2019) Identification of peripheral neural circuits that regulate heart rate using optogenetic and viral vector strategies. *Nat. Commun.* 10, 1944
63. Critchley, H.D. *et al.* (2004) Neural systems supporting interoceptive awareness. *Nat. Neurosci.* 7, 189–195
64. Malik, S. *et al.* (2018) Deep TMS of the insula using the H-coil modulates dopamine release: a crossover [^{11}C] PHNO-PET pilot trial in healthy humans. *Brain Imaging Behav.* 12, 1306–1317
65. Mayer, E.A. and Collins, S.M. (2002) Evolving pathophysiological models of functional gastrointestinal disorders. *Gastroenterology* 122, 2032–2048
66. Quigley, K.S. *et al.* (2021) Functions of interoception: from energy regulation to experience of the self. *Trends Neurosci.* 44, 29–38
67. Barrett, L.F. and Bliss-Moreau, E. (2009) Affect as a psychological primitive. *Adv. Exp. Soc. Psychol.* 41, 167–218
68. Pavlovich, L.A. and Valentino, R.J. (1995) Central regulation of micturition in the rat: the corticotropin-releasing hormone from Barrington's nucleus. *Neurosci. Lett.* 196, 185–188
69. Hou, X.H. *et al.* (2016) Central control circuit for context-dependent micturition. *Cell* 167, 73–86
70. Valentino, R.J. *et al.* (1994) Evidence for widespread afferents to Barrington's nucleus, a brainstem region rich in corticotropin-releasing hormone neurons. *Neuroscience* 62, 125–143
71. Brener, J. and Ring, C. (2016) Towards a psychophysics of interoceptive processes: the measurement of heartbeat detection. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 371, 20160015
72. Garfinkel, S.N. *et al.* (2016) Discrepancies between dimensions of interoception in autism: Implications for emotion and anxiety. *Biol. Psychol.* 114, 117–126
73. Garfinkel, S.N. and Critchley, H.D. (2016) Threat and the body: how the heart supports fear processing. *Trends Cogn. Sci.* 20, 34–46
74. D'Alonzo, M. *et al.* (2020) Modulation of body representation impacts on efferent autonomic activity. *J. Cogn. Neurosci.* 32, 1104–1116
75. Zaman, J. *et al.* (2020) The use of stimulus perception to account for variability in skin conductance responses to interoceptive stimuli. *Psychophysiology* 57, e13494
76. Mehling, W. (2016) Differentiating attention styles and regulatory aspects of self-reported interoceptive sensibility. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 371, 20160013
77. Park, J. *et al.* (2012) Abnormal sympathetic reactivity to the cold pressor test in overweight humans. *Am. J. Hypertens.* 25, 1236–1241
78. Ackley, R. and Watkins, R.H. (2018) Microneurography as a tool to study the function of individual C-fiber afferents in humans: responses from nociceptors, thermoreceptors, and mechanoreceptors. *J. Neurophysiol.* 120, 2834–2846
79. Freeman, R. and Chapleau, M.W. (2013) Testing the autonomic nervous system. *Handb. Clin. Neurol.* 115, 115–136
80. Maister, L. *et al.* (2017) Neurobehavioral evidence of interoceptive sensitivity in early infancy. *Elife* 6, e25318
81. Khalsa, S.S. *et al.* (2009) Interoceptive awareness declines with age. *Psychophysiology* 46, 1130–1136
82. Mikkelsen, M.B. *et al.* (2019) Emotional reactivity and interoceptive sensitivity: exploring the role of age. *Psychon. Bull. Rev.* 26, 1440–1448
83. Li, D. *et al.* (2017) Adolescent development of insula-dependent interoceptive regulation. *Dev. Sci.* 20, e12438
84. Toller, G. *et al.* (2018) Individual differences in socioemotional sensitivity are an index of salience network function. *Cortex* 103, 211–223
85. Maleki, N. *et al.* (2015) Female migraineurs show lack of insular thinning with age. *Pain* 156, 1232–1239
86. Suarez-Roca, H. *et al.* (2019) Contribution of baroreceptor function to pain perception and perioperative outcomes. *Anesthesiology* 130, 634–650
87. Migliorini, R. *et al.* (2013) What do you feel? Adolescent drug and alcohol users show altered brain response to pleasant interoceptive stimuli. *Drug Alcohol Depend.* 133, 661–668
88. Avery, J.A. *et al.* (2014) Major depressive disorder is associated with abnormal interoceptive activity and functional connectivity in the insula. *Biol. Psychiatry* 76, 258–266
89. Grossi, D. *et al.* (2017) Altered functional connectivity of interoception in illness anxiety disorder. *Cortex* 86, 22–32
90. Simmons, A. *et al.* (2009) Initial evidence of a failure to activate right anterior insula during affective set shifting in posttraumatic stress disorder. *Psychosom. Med.* 71, 373–377
91. Stern, E.R. *et al.* (2020) The buildup of an urge in obsessive-compulsive disorder: behavioral and neuroimaging correlates. *Hum. Brain Mapp.* 41, 1611–1625
92. Hatfield, T.R. *et al.* (2019) Autism spectrum disorder and interoception: abnormalities in global integration? *Autism* 23, 212–222
93. Donofry, S.D. *et al.* (2016) Alterations in emotion generation and regulation neurocircuitry in depression and eating disorders: a comparative review of structural and functional neuroimaging studies. *Neurosci. Biobehav. Rev.* 68, 911–927
94. Harshaw, C. (2015) Interoceptive dysfunction: toward an integrated framework for understanding somatic and affective disturbance in depression. *Psychol. Bull.* 141, 311–363
95. Garcia-Cordero, I. *et al.* (2016) Feeling, learning from and being aware of inner states: interoceptive dimensions in neurodegeneration and stroke. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 371, 20160006
96. Garcia-Cordero, I. *et al.* (2015) Stroke and neurodegeneration induce different connectivity aberrations in the insula. *Stroke* 46, 2673–2677
97. Bonaz, B. *et al.* (2021) Diseases, disorders, and comorbidities of interoception. *Trends Neurosci.* 44, 39–51
98. Elsenbruch, S. *et al.* (2010) Patients with irritable bowel syndrome have altered emotional modulation of neural responses to visceral stimuli. *Gastroenterology* 139, 1310–1319
99. Kanoski, S.E. and Grill, H.J. (2017) Hippocampus contributions to food intake control: mnemonic, neuroanatomical, and endocrine mechanisms. *Biol. Psychiatry* 81, 748–756
100. Avery, J.A. *et al.* (2017) Obesity is associated with altered mid-insula functional connectivity to limbic regions underlying appetitive responses to foods. *J. Psychopharmacol.* 31, 1475–1484
101. Ardizzi, M. *et al.* (2016) Interoception and positive symptoms in schizophrenia. *Front. Hum. Neurosci.* 10, 379
102. Kutschke, K. *et al.* (2019) Interoceptive awareness in patients with attention-deficit/hyperactivity disorder (ADHD). *Atten. Defic. Hyperact. Disord.* 11, 395–401
103. Uddin, L.Q. and Menon, V. (2009) The anterior insula in autism: under-connected and under-examined. *Neurosci. Biobehav. Rev.* 33, 1198–1203

104. Gu, X. *et al.* (2018) Heightened brain response to pain anticipation in high-functioning adults with autism spectrum disorder. *Eur. J. Neurosci.* 47, 592–601
105. Herman, A.M. and Duka, T. (2019) Facets of impulsivity and alcohol use: what role do emotions play? *Neurosci. Biobehav. Rev.* 106, 202–216
106. Goldstein, R.Z. *et al.* (2009) The neurocircuitry of impaired insight in drug addiction. *Trends Cogn. Sci.* 13, 372–380
107. Duncan, A. *et al.* (2019) Habenular TCF7L2 links nicotine addiction to diabetes. *Nature* 574, 372–377
108. Farokhnia, M. *et al.* (2019) Ghrelin: From a gut hormone to a potential therapeutic target for alcohol use disorder. *Physiol. Behav.* 204, 49–57
109. Morris, L.S. *et al.* (2018) Stress, motivation, and the gut–brain axis: a focus on the ghrelin system and alcohol use disorder. *Alcohol. Clin. Exp. Res.*
110. Suarez, A.N. *et al.* (2019) Regulation of memory function by feeding-relevant biological systems: following the breadcrumbs to the hippocampus. *Front. Mol. Neurosci.* 12, 101
111. Weng, H.Y. *et al.* (2021) Interventions and manipulations of interoception. *Trends Neurosci.* 44, 52–62
112. Barrera, T.L. *et al.* (2014) A review of cognitive behavioral therapy for panic disorder in patients with chronic obstructive pulmonary disease: the rationale for interoceptive exposure. *J. Clin. Psychol. Med. Settings* 21, 144–154
113. Freeman, S.C. *et al.* (2018) Component network meta-analysis identifies the most effective components of psychological preparation for adults undergoing surgery under general anesthesia. *J. Clin. Epidemiol.* 98, 105–116
114. Ehde, D.M. *et al.* (2014) Cognitive-behavioral therapy for individuals with chronic pain: efficacy, innovations, and directions for research. *Am. Psychol.* 69, 153–166
115. Moisset, X. *et al.* (2020) Neurostimulation methods in the treatment of chronic pain. *J. Neural Transm. (Vienna)* 127, 673–686
116. Seth, A.K. *et al.* (2011) An interoceptive predictive coding model of conscious presence. *Front. Psychol.* 2, 395
117. Owens, A.P. *et al.* (2018) Interoceptive inference: from computational neuroscience to clinic. *Neurosci. Biobehav. Rev.* 90, 174–183
118. Di Lemia, D. *et al.* (2018) Feel the time. Time perception as a function of interoceptive processing. *Front. Hum. Neurosci.* 12, 74
119. Petzschner, F.H. *et al.* (2021) Computational models of interoception and body regulation. *Trends Neurosci.* 44, 63–76